Training-Free Safety Guidance for Action-Space Diffusion Policies via Tweedie-Adaptive Constrained Sampling

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by

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List of Acronyms

CBF Control Barrier Function. A scalar function whose sublevel set defines the safe region; used with a forward-invariance condition to keep a dynamical system safe.

DDIM Denoising Diffusion Implicit Model. A deterministic accelerated sampler for diffusion models that enables fewer-step generation.

DP Diffusion Policy. A visuomotor control paradigm that formulates action prediction as conditional denoising over action chunks.

EE End-Effector. The tool frame of a robotic manipulator (typically gripper or hand).

PF Potential Field. A classical reactive obstacle-avoidance method using attractive/repulsive gradient fields.

QP Quadratic Program. An optimization problem with quadratic objective and linear constraints.

TACS Tweedie-Adaptive Constrained Sampling. The training-free inference-time safety method proposed in this thesis.

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Abstract of the Thesis

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Diffusion policies have become the dominant paradigm for learned robotic manipulation, yet they offer no intrinsic mechanism to enforce workspace safety constraints. Existing inference-time safety methods target *state-space* trajectory planners, where explicit waypoints can be constrained directly; extending such guidance to *action-space* diffusion policies is non-trivial because the generated trajectory contains robot actions, not states. This thesis presents **TACS** (Tweedie-Adaptive Constrained Sampling), a training-free inference-time safety layer for action-space diffusion policies. At each DDIM denoising step, TACS evaluates a barrier-inspired potential on the Tweedie-predicted end-effector position and applies a repulsive correction to the predicted clean action, letting subsequent denoising steps coherently adapt the full trajectory. On three robosuite manipulation tasks (Lift, Can, Square), TACS achieves consistently better success–violation trade-offs than post-hoc potential-field projection, with a statistically significant win–win on Lift (p=0.015) and no measurable inference overhead. An adaptive guidance schedule $s(\tau)$ is proposed and shown through ablation to be dominated by the spatial self-gating of the guidance weight, making the simple constant schedule the recommended choice.

Chapter 1

Introduction

Diffusion policies [Chi et al.(2023)] have become the dominant paradigm for learned robotic manipulation [Ze et al.(2024), Liu et al.(2025), Zhao et al.(2023)], generating robot actions as conditional denoising over high-dimensional action chunks [Ho et al.(2020), Sohl-Dickstein et al.(2015)]. Enabled by large-scale datasets [Open X-Embodiment(2024), Mandlekar et al.(2021)], they achieve state-of-the-art performance across diverse tasks. However, these policies provide no intrinsic mechanism to enforce workspace safety constraints: a trained policy may generate trajectories that collide with obstacles, violate joint limits, or exceed velocity bounds.

Recent work on safe diffusion addresses this gap for *state-space trajectory planners* [Janner et al.(2022)]—SafeDiffuser [Xiao et al.(2025)] solves CBF-QPs per denoising step, CoDiG [Ma et al.(2025)] augments the score with barrier gradients, and Constrained Diffusers [Zhang et al.(2025)] applies constrained Langevin sampling. However, these methods generate reference trajectories in state space, tracked by a separate controller. They are not directly applicable to *action-space* diffusion policies [Chi et al.(2023)], which produce robot actions directly as high-dimensional chunks (e.g., 10D: position, rotation, and gripper) conditioned on observations as a global vector. This architectural difference creates three challenges for inference-time safety: (1) safety constraints must be evaluated on task-relevant dimensions extracted from the predicted actions, not from explicit states; (2) observations condition the model globally and are absent from the generated trajectory, so guidance cannot operate on them; and (3) guidance targets only a subset of action dimensions (3 of 10 for EE position), while the remaining dimensions must adapt coherently through the diffusion process.

We propose **TACS** (Tweedie-Adaptive Constrained Sampling), a training-free method

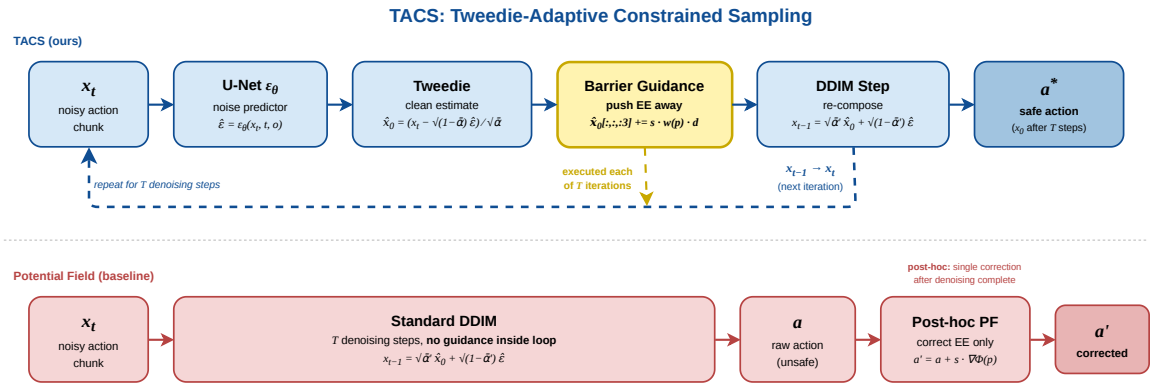


Figure 1.1: **TACS vs. post-hoc PF projection.** *Top (TACS, ours):* the gold-highlighted BARRIER GUIDANCE block is the *only* algorithmic intervention; the gold arrow marks that it fires inside every one of T DDIM iterations, while the blue dashed arc closes the loop. *Bottom (PF baseline):* T unguided DDIM steps followed by a single post-hoc correction $a'{=}a + s\,\nabla\Phi(p)$ on the final action. Algorithmic details: pipeline in Section 4, why it wins in Section 5.5.

that injects barrier-inspired guidance directly into the DDIM denoising loop of action-space diffusion policies (Figure 1.1). At each denoising step, TACS computes the Tweedie-predicted clean trajectory $\hat{x}_0$ [Efron(2011), Song et al.(2021b)], applies repulsive guidance on the end-effector position dimensions, and lets subsequent denoising steps adapt the full trajectory—including rotation and gripper—to be consistent with the safety correction. This *in-loop* guidance compounds across denoising steps, achieving effective obstacle avoidance at $\sim$10$\times$ smaller guidance scales than post-hoc projection. Our contributions:

- We introduce **TACS**, an inference-time barrier-inspired guidance method for action-space diffusion policies, operating on Tweedie-predicted end-effector positions within the denoising loop;
- We propose an **adaptive guidance schedule** $s(\tau)$ motivated by the improving signal-to-noise ratio of Tweedie predictions [Shen et al.(2024)] throughout denoising;
- We demonstrate consistently **better success-violation trade-offs** than post-hoc potential-field projection [Khatib(1986)] on all three robosuite [Robosuite(2020)] manipulation tasks, with a statistically significant win-win on Lift ($p{=}0.015$, 5 seeds) and no measurable inference overhead.

Chapter 2

Related Work

Safety in diffusion-based robotics has so far been studied almost exclusively for *state-space trajectory planners*, where the generated sequence contains explicit waypoints on which barrier functions can be evaluated directly. This line of work has produced a rich algorithmic toolkit (CBF-QP, barrier-gradient score augmentation, predict–project–renoise, constrained Langevin) but leaves open the question of how to inject safety constraints into *action-space* policies, whose generated trajectories contain raw robot actions instead of states and whose observations enter only as a global conditioning vector. A second, largely disjoint thread — test-time guidance for action-space policies — has emerged for *reward* maximization (Q-value guidance, score composition) but not for safety. Table 2.1 positions TACS along these two axes: we evaluate each method by its *target architecture* (SP vs. AP), its *objective* (safety vs. reward), its *guidance mechanism*, and whether it is training-free, manipulation-validated, and low-overhead. TACS is the only entry combining inference-time safety guidance with action-space diffusion policies for manipulation.

Safe diffusion for robotics. SafeDiffuser [Xiao et al.(2025)] embeds CBFs [Ames et al.(2017)] into the diffusion reverse process via per-step QP, providing formal safety guarantees for state-space trajectory planners [Janner et al.(2022)] at the cost of significant latency. CoDiG [Ma et al.(2025)] reduces this overhead by augmenting the denoising score with barrier gradients, achieving real-time performance for 2D autonomous racing in Frenet coordinates. SafeFlow [SafeFlow(2025)] and SafeFlowMatcher [Jang(2025)] extend CBF guidance to flow matching [Lipman et al.(2023)]. PPR [Rochman-Sharabi and Louppe(2026)] alternates prediction, projection, and renoising for hard constraints. Constrained Diffusers [Zhang et al.(2025)] applies constrained Langevin sampling with discrete CBFs. SafeBimanual [Deng et al.(2025)] guides bimanual trajectories, EB-

Table 2.1: **Positioning among safe-/test-time-guided diffusion methods.** Target: SP = state-space planner, AP = action-space policy; (FM) denotes flow-matching variants. Goal: Safety (obstacle / barrier), Reward (return maximization). TF = training-free. Manip = validated on visuomotor manipulation. Overhead relative to unmodified inference; "nr" = not reported. Selection criterion: methods that explicitly modify the denoising/flow trajectory for safety or reward; pure architectural extensions are discussed in text.

Method	Target	Goal	Guidance mechanism	TF	Manip	Overhead
SafeDiffuser [Xiao et al.(2025)]	SP	Safety	CBF-QP per step	✓	$\times^{\dagger}$	$\sim$473 ms
CoDiG [Ma et al.(2025)]	SP	Safety	Score + ∇B	✓	$\times^{\ddagger}$	$\sim$20 ms
SafeFlow [SafeFlow(2025)]	SP (FM)	Safety	CBF-QP on FM	✓	×	nr
SafeFlowMatcher [Jang(2025)]	SP (FM)	Safety	Predict-correct	✓	×	nr
Constrained Diff. [Zhang et al.(2025)]	SP	Safety	Constrained Langevin	✓	×	nr
PPR [Rochman-Sharabi and Louppe(2026)]	SP	Safety	Predict-Project-Renoise	✓	×	nr
DualShield [Yang et al.(2026)]	SP	Safety	Reachability + guidance	✓	×	nr
EB-MBD [Mishra et al.(2026)]	SP	Safety	Emerging barrier fns	✓	×	nr
SafeBimanual [Deng et al.(2025)]	SP (bimanual)	Safety	Guided sampling	✓	✓*	nr
PRESTO [Seo et al.(2025)]	SP	Safety	Hinge loss (training)	×	×	—
VGD [Ye et al.(2025)]	AP	Reward	Q-value guidance	✓	✓	≈vanilla
TACS (ours)	**AP**	**Safety**	**Tweedie $\hat{x}_0$ barrier**	✓	✓	≈vanilla

† Evaluated on Walker2D/Hopper/Maze, not visuomotor manipulation. ‡ Autonomous racing. * Bimanual collision only.

MBD [Mishra et al.(2026)] introduces emerging barriers, and DualShield [Yang et al.(2026)] combines reachability with diffusion for driving. A common thread among these safety-constrained diffusion methods: *all* operate on state-space trajectory planners where the generated sequence contains explicit states. Prior inference-time safety methods thus target SP planners; extending safety guidance to action-space policies, where the trajectory contains robot actions directly, remains largely unexplored.

Action-space diffusion policies. Diffusion Policy [Chi et al.(2023)] formulates visuomotor control as conditional denoising over action chunks with observations as global context, spawning a rapidly growing family: score-based variants (BESO) [Reuss et al.(2023)], 3D-aware extensions [Ze et al.(2024)], one-step distillation [Prasad et al.(2024)], RL fine-tuning [Ren et al.(2025)], and foundation-scale models [Liu et al.(2025)]. Recent work explores test-time modification of these policies: VGD [Ye et al.(2025)] steers frozen policies via Q-function gradients for reward maximization, D3P [Yu et al.(2025)] dynamically adjusts denoising steps, GPC [GPC(2025)] composes policy scores, and contractive policies [Abyaneh et al.(2026)] improve sampling robustness. For

motion planning diffusion models (distinct from action-space policies), PRESTO [Seo et al.(2025)] adds hinge-loss collision terms during training, AdaptDiffuser [Liang et al.(2023)] enables self-evolving planning, and SE(3)-DiffusionFields [Urain et al.(2023)] operate in task space. None of these methods address inference-time *safety constraints* for action-space policies. TACS fills this gap.

Guided diffusion sampling. Classifier guidance [Dhariwal and Nichol(2021)] and classifier-free guidance [Ho and Salimans(2022)] modify the denoising score [Song et al.(2021a)] to steer generation. TFG [Ye et al.(2024)] provides a unified framework for training-free guidance, while CFG++ [Chung et al.(2025)] addresses off-manifold artifacts. For inverse problems, DPS [Chung et al.(2023)] and Universal Guidance [Bansal et al.(2023)] enable conditioning without retraining; RePaint [Lugmayr et al.(2022)] uses iterative renoising for inpainting. In the $\hat{x}_0$-prediction paradigm central to TACS, DRaFT [Clark et al.(2024)] fine-tunes diffusion models via differentiable rewards on Tweedie predictions, and PIDM [Bastek et al.(2025)] enforces physics losses on $\hat{x}_0$. Test-time alignment [Kim et al.(2025)] applies SMC-based guidance without reward over-optimization. Critically, Shen et al. [Shen et al.(2024)] show that training-free Tweedie guidance suffers from misaligned gradients at high noise levels, directly motivating our adaptive schedule $s(\tau)$ that reduces guidance in early (noisy) denoising steps.

Safety primitives. Our work builds on two classical foundations: artificial potential fields [Khatib(1986)], which define repulsive forces for obstacle avoidance (the post-hoc PF baseline), and control barrier functions [Ames et al.(2017), Ames et al.(2019)], which provide forward invariance guarantees. The safety layer of Dalal et al. [Dalal et al.(2018)]—a differentiable action-correction module atop a policy network—is a closely related action-space safety filter; TACS instead operates at the denoising stage before actions are committed. Recovery RL [Thananjeyan et al.(2021)] and recent surveys [Garg et al.(2023)] provide broader context on safe robot learning. TACS bridges these safety primitives with the diffusion generative process, applying barrier-inspired guidance inside the denoising loop rather than as a post-hoc filter.

Chapter 3

Preliminaries

3.1 Action-Space Diffusion Policy

Diffusion models [Sohl-Dickstein et al.(2015), Ho et al.(2020)] learn to reverse a Gaussian noising process, generating samples via iterative denoising. Score-based formulations [Song et al.(2021a)] unify this with score matching. Following Chi et al. [Chi et al.(2023)], we model a visuomotor policy as a conditional denoising diffusion process over action chunks $a \in \mathbb{R}^{H \times D_a}$, where H is the prediction horizon and D_a the action dimension. Unlike state-space trajectory planners [Janner et al.(2022), Ajay et al.(2023)] where the diffusion operates over state-action sequences, action-space policies condition on the observation history $o_{0:T_o}$ as a global vector (FiLM conditioning), *not* as part of the denoised trajectory. Training minimizes:

$$\mathcal{L}_{\text{denoise}} = \mathbb{E}_{a_0 \sim \mathcal{D},\, \epsilon \sim \mathcal{N}(\mathbf{0},\mathbf{I}),\, t \sim \mathcal{U}\{1,T\}} \left[\|\epsilon_\theta(a_t, t, o) - \epsilon\|^2 \right] \tag{3.1}$$

where $a_t = \sqrt{\bar{\alpha}_t}\, a_0 + \sqrt{1 - \bar{\alpha}_t}\, \epsilon$ is the noised action, $\bar{\alpha}_t \in (0, 1]$ is the cumulative noise schedule ($\bar{\alpha}_0 \approx 1$, $\bar{\alpha}_T \approx 0$), and ϵ_θ is a 1D temporal U-Net trained to predict the added noise. We use x_t to denote the noised action trajectory at diffusion step t interchangeably with a_t.

3.2 DDIM Sampling and Tweedie's Formula

At inference, deterministic DDIM [Song et al.(2021b)] (η=0) generates actions by iterating:

$$\hat{x}_0 = \frac{x_t - \sqrt{1 - \bar{\alpha}_t}\, \epsilon_\theta(x_t, t, o)}{\sqrt{\bar{\alpha}_t}} \tag{3.2}$$

$$x_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\,\hat{x}_0 + \sqrt{1-\bar{\alpha}_{t-1}}\,\epsilon_\theta(x_t, t, o) \tag{3.3}$$

Eq. 3.2 follows from algebraically inverting the forward noising $a_t = \sqrt{\bar{\alpha}_t}\,a_0 + \sqrt{1-\bar{\alpha}_t}\,\epsilon$ using the network's noise estimate $\hat{\epsilon} = \epsilon_\theta(x_t, t, o)$; Eq. 3.3 is the deterministic DDIM update recombining $\hat{x}_0$ with $\hat{\epsilon}$ to produce the next iterate x_{t-1}. The quantity $\hat{x}_0$ equals the posterior mean $\mathbb{E}[a_0 \mid a_t]$ under the Gaussian diffusion model—Tweedie's formula in the empirical Bayes literature [Efron(2011)]—and its quality improves monotonically as $t \to 0$. Prior work has exploited this property for physics losses [Bastek et al.(2025)] and reward fine-tuning [Clark et al.(2024)]; it is also the foundation on which TACS operates (Section 4).

3.3 Barrier Functions

A barrier function $B : \mathcal{X} \to \mathbb{R}$ is a scalar safety score whose super-level set defines the safe region $\mathcal{C} = \{x \in \mathcal{X} : B(x) \geq 0\}$. Given a control-affine system $\dot{x} = f(x) + g(x)\,u$, B is a *control barrier function* (CBF) [Ames et al.(2017), Ames et al.(2019)] if there exists a class-$\mathcal{K}$ function α such that for all $x \in \mathcal{X}$:

$$\sup_u [\, L_f B(x) + L_g B(x)\,u + \alpha(B(x))\,] \geq 0, \tag{3.4}$$

where $L_f B = \nabla B \cdot f$ and $L_g B = \nabla B \cdot g$ are the Lie derivatives of B along the drift and control vector fields. *Intuition.* The time derivative of B along the system trajectory is $\dot{B} = L_f B + L_g B \cdot u$; the CBF condition asks whether there exists a feasible control u such that $\dot{B} \geq -\alpha(B)$ everywhere. Since α is a class-$\mathcal{K}$ function ($\alpha(0)$=0, strictly increasing), the inequality allows B to shrink only when the state is deep inside the safe set and forbids any decrease at the boundary B=0. Satisfying Eq. 3.4 guarantees *forward invariance* of $\mathcal{C}$: the state, once inside, provably never leaves.

At runtime, a CBF is enforced by projecting a nominal control u_{nom} (from a policy or planner) onto the safety constraint via a quadratic program:

$$u^\star = \arg\min_u \|u - u_{\text{nom}}\|^2 \quad \text{s.t.} \quad L_f B + L_g B \cdot u + \alpha(B) \geq 0. \tag{3.5}$$

The CBF-QP must be solved at *every* control step, and when embedded inside a T-step diffusion loop (as in SafeDiffuser [Xiao et al.(2025)] and SafeFlow [SafeFlow(2025)]) its cost multiplies by T, typically giving hundreds of milliseconds per trajectory (Table 2.1).

For spherical obstacle avoidance with point $p \in \mathbb{R}^3$, obstacle center c and radius r, a canonical barrier function is:

$$B(p) = \|p - c\| - (r + \delta) \tag{3.6}$$

where $\delta > 0$ is a safety margin. Concretely, $B(p)$ equals the signed distance from p to the inflated obstacle surface: positive in free space, zero at the safety-margin boundary, negative inside the obstacle zone. We define a continuous guidance weight in Section 4.

Chapter 4

Method: TACS

4.1 Problem Formulation

Consider a pre-trained action-space diffusion policy ϵ_θ that generates action chunks $a \in \mathbb{R}^{H \times D_a}$ conditioned on observations o. Each action chunk contains H horizon steps.

Action-space assumption. We assume the policy operates in *absolute end-effector* action mode: the first three dimensions $a[h, :3]$ directly represent the EE Cartesian position $p_h \in \mathbb{R}^3$ at horizon step h. This matches the absolute-pose variant of Diffusion Policy [Chi et al.(2023)] used in our experiments. Extensions to delta-action (requires integrating Δp over the horizon) and joint-space action (requires differentiable forward kinematics to map $q \mapsto p$) are discussed in Section 6.1.

Given a set of workspace obstacles $\{(c_j, r_j)\}_{j=1}^{M}$ with centers c_j and radii r_j, we define safety via barrier functions $B_j(p) = \|p - c_j\| - (r_j + \delta) \geq 0$ (Eq. 3.6). Our goal is to modify the DDIM sampling process such that the generated EE positions satisfy $B_j(p_h) \geq 0$ for all h, j, *without retraining* the policy and with *negligible computational overhead.*

Design choice: skip the CBF-QP. The standard route to enforce Eq. 3.4 at runtime is the CBF-QP of Eq. 3.5, embedded in the denoising loop [Xiao et al.(2025), SafeFlow(2025)]; this path grants formal forward-invariance but pays the per-step QP cost. TACS takes the opposite trade-off: we use B only as a signed-distance safety score that modulates the denoising process, with no QP solve, no control-affine dynamics assumption, and no runtime satisfaction of Eq. 3.4. Concretely, the generic state x in Eq. 3.4 is replaced by the predicted end-effector position p_h, reducing the safety check to a pure geometric query $B_j(p_h)$. We forfeit the formal forward-invariance guarantee in exchange for

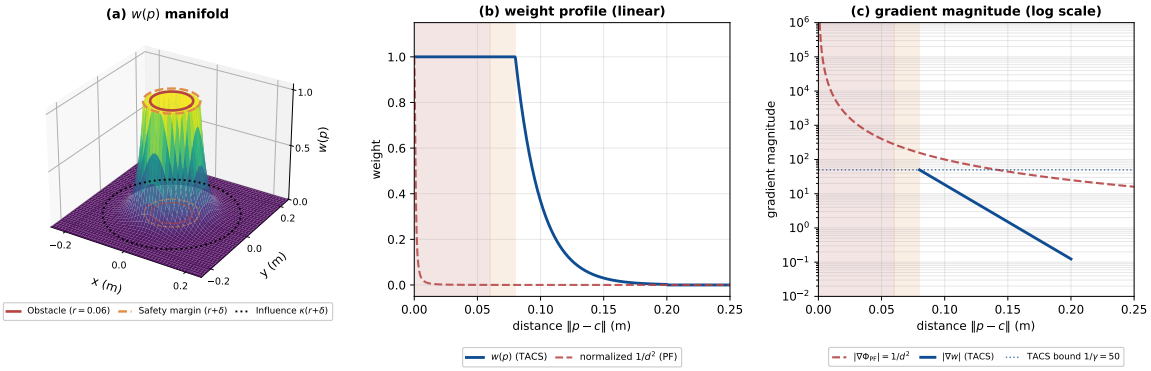


Figure 4.1: **Guidance weight $w(p)$ and its gradient** (Eq. 4.1; Lift parameters r=0.06, δ=0.02, γ=0.02 m, κ=2.5). *(a)* 3D manifold z=$w(p)$; note that the plateau extends over both the safety margin *and* the obstacle interior—the $\max(B, 0)$ clamp saturates w to 1 whenever unsafe. *(b)* Radial profile of w vs. normalized inverse-distance PF. *(c)* Gradient magnitude on log scale: $\|\nabla w\|$ bounded by $1/\gamma$=50, vs. $1/d^2$ divergence of the PF force.

vanilla-DDIM latency on visuomotor policies (quantified in Section 5); empirically, this soft guidance achieves strictly better success-violation Pareto fronts than the post-hoc PF baseline (Section 5.2).

4.2 Tweedie-Adaptive Constrained Sampling

As illustrated in Figure 1.1, TACS applies barrier guidance *inside* the DDIM denoising loop, so that each nudge on $\hat{x}_0$ is propagated through subsequent denoising steps; PF applies a single correction *after* denoising completes, without opportunity for the policy to adapt non-guided dimensions.

At each DDIM denoising step t, TACS:

1. Computes the Tweedie prediction $\hat{x}_0$ (Eq. 3.2)
2. Extracts predicted end-effector positions $p_h = \hat{x}_0[h, :3]$ for each horizon step h
3. Evaluates per-obstacle barrier weights $w_j(p_h)$ (Eq. 4.1) and repulsive directions $d_{h,j} = (p_h - c_j)/\|p_h - c_j\|$
4. Applies guided modification: $\hat{x}_0[h, :3] \leftarrow \hat{x}_0[h, :3] + s(\tau) \cdot \sum_j w_j(p_h)\, d_{h,j}$
5. Continues DDIM with the modified $\hat{x}_0$

Algorithm 1 gives the complete procedure with normalization and clamp steps spelled out.

The guidance weight $w(p_h)$ concentrates effort near the constraint boundary using a continuous exponential form:

$$w(p) = \exp\left(-\frac{\max(B(p), 0)}{\gamma}\right) \cdot \mathbf{1}[\|p - c\| < \kappa(r + \delta)] \tag{4.1}$$

with decay rate $\gamma > 0$ and influence radius factor $\kappa > 1$. *Intuition.* The weight $w(p)$ saturates at 1 inside the obstacle (since $\max(B, 0){=}0$ when $B{<}0$), decays exponentially as p moves away from the boundary through the influence annulus, and is hard-cut to 0 once p is beyond $\kappa(r + \delta)$ from the obstacle center. In words: *full push when unsafe, tapered push near the boundary, no push when already far away*. Figure 4.1 visualizes the resulting field. Panel (a) renders $z{=}w(p)$ as a 3D manifold: a plateau over unsafe space, an exponential cliff through the influence annulus, and a hard cutoff outside. Panel (b) plots the radial profile of w alongside a normalized inverse-distance PF for direct comparison. Panel (c) shows the gradient magnitude on log scale. Two properties matter for the rest of the paper: (i) *within the influence annulus* $r + \delta < \|p - c\| < \kappa(r + \delta)$, $\|\nabla w\|$ is bounded above by $1/\gamma{=}50$ (panel c)—unlike classical potential fields [Khatib(1986)] whose $1/d^2$ forces diverge at the obstacle surface—which gives numerical stability. At the outer cutoff $\|p - c\| = \kappa(r + \delta)$ the indicator introduces a small discontinuity (a drop from $e^{-(\kappa-1)(r+\delta)/\gamma}$ to 0), which is numerically negligible for our parameters ($e^{-6} \approx 0.0025$) and can be smoothed with a sigmoid cutoff if desired. (ii) The indicator mask sets $w{=}0$ whenever $\hat{x}_0$ lies outside the influence radius, which makes the field *spatially self-gating* and lets us dispense with temporal schedules (Section 5.4).

Key insight: Because TACS modifies $\hat{x}_0$ *inside* the denoising loop, subsequent denoising steps allow the diffusion model to *adapt the entire trajectory*—including rotation and gripper

dimensions—to be consistent with the shifted end-effector positions. Post-hoc methods (potential field projection) modify only the final output without this adaptation, leading to worse success-violation trade-offs.

Algorithm 1 TACS: Tweedie-Adaptive Constrained Sampling

Require: Policy ϵ_θ, observation o, obstacles $\{(c_j, r_j)\}$, schedule $s(\cdot)$, normalizer ϕ, total steps N

$x_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ // Sample Gaussian noise
for $i = 0, 1, \ldots, N-1$ **do**
  $t \leftarrow \text{timesteps}[i]$
  $\hat{\epsilon} \leftarrow \epsilon_\theta(x_t, t, o)$ // Predict noise
  $\hat{x}_0 \leftarrow (x_t - \sqrt{1-\bar{\alpha}_t}\,\hat{\epsilon})/\sqrt{\bar{\alpha}_t}$ // Tweedie prediction (Eq. 3.2)
  $\hat{x}_0 \leftarrow \text{clamp}(\hat{x}_0, -1, 1)$ // Enforce normalized action range
  $\hat{a} \leftarrow \phi^{-1}(\hat{x}_0)$ // Unnormalize to physical action space
  for each horizon step $h = 0, \ldots, H-1$ **do**
    $\hat{p}_h \leftarrow \hat{a}[h, :3]$ // Predicted EE position
    for each obstacle $j = 1, \ldots, M$ **do**
      $d_{h,j} \leftarrow (\hat{p}_h - c_j)/\|\hat{p}_h - c_j\|$, $w_{h,j} \leftarrow$ Eq. (4.1)
      $\hat{a}[h, :3] \mathrel{+}= s(\tau_i) \cdot w_{h,j} \cdot d_{h,j}$ **// Barrier guidance**
    end for
  end for
  $\hat{x}_0 \leftarrow \text{clamp}(\phi(\hat{a}),\, -1,\, 1)$ // Re-normalize and clamp
  $x_{t-1} \leftarrow \sqrt{\bar{\alpha}_{t-1}}\,\hat{x}_0 + \sqrt{1-\bar{\alpha}_{t-1}}\,\hat{\epsilon}$ // DDIM step (Eq. 3.3)
end for
return $\phi^{-1}(x_0)$ as action trajectory // $x_0 \equiv x_t$ after N steps

4.3 Adaptive Guidance Schedule

The quality of $\hat{x}_0$ varies substantially across denoising steps: at $t \approx T$ (early denoising), $\hat{x}_0$ is dominated by noise and barrier guidance on it is unreliable; at $t \approx 0$ (late denoising), $\hat{x}_0$ closely approximates the clean trajectory. We propose a time-dependent guidance scale:

$$s(\tau) = s_{\text{base}} \cdot \sigma(\tau) \cdot \mathbf{1}[\tau \geq \tau_{\text{wu}}] \tag{4.2}$$

where $\sigma : [0, 1] \to [0, 1]$ is a monotonically increasing function of the denoising progress $\tau = i/N \in [0, 1]$ (with i the step index and N the total steps; τ=0 at the noisiest step, τ=1 at the cleanest), and

$\tau_{\text{wu}} = 0.1$ is a warmup threshold that disables guidance during the noisiest 10% of denoising steps where $\hat{x}_0$ is unreliable. We evaluate three schedules: linear $\sigma(\tau) = \tau$, cosine $\sigma(\tau) = \frac{1}{2}(1-\cos(\pi\tau))$, and $\sqrt{\cdot}$ schedule $\sigma(\tau) = \sqrt{\tau}$. *Intuition.* The schedule encodes a *trust-the-prediction* principle: when the denoiser still produces noisy $\hat{x}_0$ (early steps), barrier guidance is acting on unreliable position estimates, so we hold back; as $\hat{x}_0$ becomes a reliable preview of the final trajectory (late steps), we turn guidance up.

Consistency of $\hat{x}_0$ and $\hat{\epsilon}$. Algorithm 1 modifies $\hat{x}_0$ by a barrier correction $\Delta\hat{x}_0 = s(\tau)\sum_j w_j d_j$ and then applies the standard DDIM step $x_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\,\hat{x}_0^{\text{mod}} + \sqrt{1-\bar{\alpha}_{t-1}}\,\hat{\epsilon}$ using the *original* noise estimate $\hat{\epsilon}$, rather than re-deriving $\hat{\epsilon}_{\text{mod}} = (x_t - \sqrt{\bar{\alpha}_t}\,\hat{x}_0^{\text{mod}})/\sqrt{1-\bar{\alpha}_t}$ as strict consistency with Eq. 3.2 would require. This is a deliberate design choice consistent with CFG++ [Chung et al.(2025)] and with loss-guidance methods [Bansal et al.(2023), Shen et al.(2024)]: (i) the per-horizon-step correction is bounded by $\|\Delta\hat{x}_0[h,:3]\|_2 \le s_{\text{base}}\,\sigma(\tau)\,M$ (since $w_j \le 1$ and $\|d_{h,j}\|$=1), and is further suppressed by the warmup that disables guidance during the noisiest steps; the resulting first-order approximation error in the subsequent DDIM update is $O(\|\Delta\hat{x}_0\|^2)$; (ii) updating only $\hat{x}_0$ preserves the off-manifold stability that motivates CFG++. A strict $\hat{\epsilon}$-resync variant is a natural ablation; we leave its empirical comparison to future work.

Chapter 5

Experiments

5.1 Setup

Tasks. We evaluate on three robosuite [Robosuite(2020)] manipulation tasks (task scenes shown in Figure 5.2), each augmented with a single spherical obstacle (safety margin δ=0.02m): **Lift** (transit obstacle at center $(0.08, 0.04, 0.88)$, radius 0.06m), **Can** (approach-path obstacle at $(0.15, 0.05, 1.02)$, radius 0.10m), and **Square** (workspace-center obstacle at $(0.00, 0.12, 0.95)$, radius 0.10m). Each task uses a pre-trained Diffusion Policy [Chi et al.(2023)] checkpoint with absolute action mode.

Baselines.

- **Vanilla**: unmodified Diffusion Policy (no safety mechanism)
- **PF** (Potential Field): post-hoc repulsive projection applied to the final action trajectory after denoising completes
- **TACS** (ours): barrier guidance inside DDIM loop (Algorithm 1)

Metrics. Success rate (%), constraint violation rate (% of simulation steps where any end-effector-to-obstacle distance falls below $r + \delta$), and inference speed (iterations/sec). All results report mean $\pm$ standard deviation across 3 random seeds $\times$ 50 episodes per seed. Statistical significance is assessed via Welch's t-test where noted.

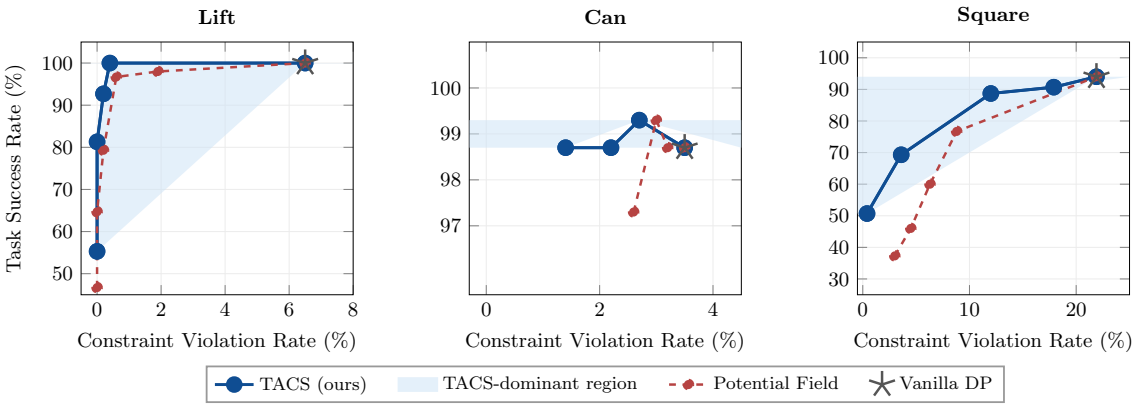


Figure 5.1: **Success-vs-violation Pareto fronts.** Each marker is one guidance-scale setting; curves trace the full s sweep (TACS: large blue dots; PF: small red dots; Vanilla: gray star). Shaded region = operating points Pareto-dominated by TACS. On all three tasks the TACS frontier lies strictly to the upper-left of the PF frontier. *Axis scales differ across panels.* 3 seeds $\times$ 50 episodes per point.

Hyperparameters. Barrier parameters: δ=0.02, γ=0.02, influence factor κ=2.5. Guidance scale selected per task from a Pareto sweep (Section 5.2).

5.2 Main Results: Pareto Analysis

Figure 5.1 shows the success-vs-violation Pareto front for TACS and PF across guidance scales. On all three tasks, TACS achieves strictly better trade-offs:

Lift: Win-win. TACS at s=0.005 maintains 100% success while reducing violations from 6.5% to 0.4%—a 16$\times$ reduction. PF requires $s\geq$0.02 to reach comparable violation levels, at which point success drops to 96.7%.

Can: Near win-win. TACS at s=0.02 preserves baseline success (98.7%) while reducing violations by 60% (3.5% $\rightarrow$ 1.4%). PF at comparable violation requires s=0.1, losing 1.4% success.

Square: Pareto-dominant. Square is the most constrained task, with the obstacle occupying the workspace center, and both methods incur substantial trade-offs here: TACS at s=0.01 retains 88.7% success with 12.0% violation, while PF at s=0.02 drops to 76.7% success for a lower 8.8% violation. Crucially, the Pareto front (Figure 5.1, right) shows that the *entire* PF frontier sits to the lower-right

Table 5.1: Main results at recommended guidance scales. Mean ± std across 3 seeds × 50 episodes. All methods run at the same inference speed (4.27 it/s), confirming TACS adds no measurable overhead. Best safety at matched or higher success in **bold**.

Task	Method	Success%	Violation%	it/s
Lift	Vanilla	100.0 ± 0.0	6.5 ± 3.4	4.27
	PF (s=0.02)	96.7 ± 0.9	0.6 ± 0.1	4.27
	TACS (s=0.005)	$\mathbf{100.0 \pm 0.0}$	$\mathbf{0.4 \pm 0.1}$	4.27
Can	Vanilla	98.7 ± 1.9	3.5 ± 0.5	4.27
	PF (s=0.1)	97.3 ± 1.9	2.6 ± 0.2	4.27
	TACS (s=0.02)	$\mathbf{98.7 \pm 0.9}$	$\mathbf{1.4 \pm 0.0}$	4.27
Square	Vanilla	94.0 ± 3.3	21.9 ± 3.0	4.27
	PF (s=0.02)	76.7 ± 3.4	8.8 ± 1.2	4.27
	TACS (s=0.01)	$\mathbf{88.7 \pm 2.5}$	$\mathbf{12.0 \pm 2.5}$	4.27

of TACS: even PF's lowest-violation point $(3.0\%, 37.3\%)$ is strictly dominated by the TACS point $(0.4\%, 50.7\%)$—lower violation *and* higher success. The qualitative Pareto win therefore holds across the full explored range.



Figure 5.2: **Qualitative rollouts across all three tasks, methods, and three camera views.** Each row triplet shows one task (Lift / Can / Square) from *sideback_high*, *sideback_low*, and *front (agentview)*. Columns show three key stages: *start* (pre-motion), *interaction* (gripper near the obstacle), and *task complete*. Colored trajectory overlays trace the EE path (gray = Vanilla DP, blue = TACS, red = PF); red $\times$ markers flag violation frames. Bottom-left badges on the front view report per-rollout violation counts and task success, matching Table 5.1's aggregate trends: **TACS is the only method achieving** 0 **violations with** 100% **success on all three tasks**; Vanilla incurs heavy violations $(40/2/30)$; PF either still violates (Lift 3, Can 2) or, absent on this seed but visible in Table 5.1, loses success.

5.3 Statistical Validation

To rigorously validate the Lift win-win result, we extend to 5 random seeds $\times$ 50 episodes (250 episodes total). Figure 5.3 shows the per-seed violation distributions: TACS achieves $0.32\% \pm 0.15\%$ versus Vanilla's $6.60\% \pm 3.43\%$—a $20\times$ reduction in mean with a $23\times$ reduction in seed-to-seed spread. Welch's t-test yields p=0.015 on violation (significant at α=0.05) and p=0.306 on success (non-significant), so TACS reduces violations without degrading task performance. Extended validation for Can and Square is left for the camera-ready; 3-seed results already show consistent Pareto improvements across all tasks (Table 5.1).

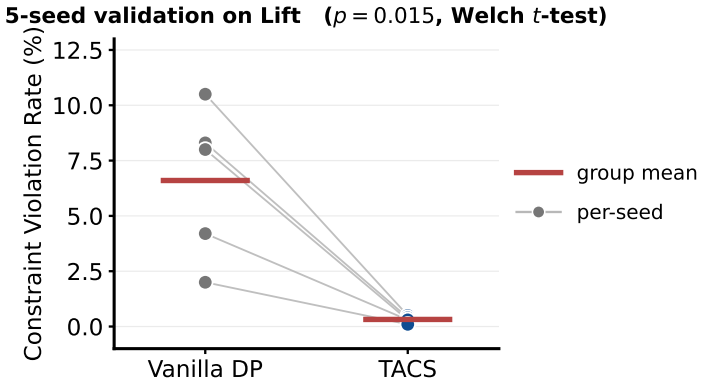


Figure 5.3: **5-seed validation on Lift** (s=0.005; 50 episodes per seed, 250 total). TACS cuts mean violation from 6.60% to 0.32% and tightens the spread (σ) from 3.43 to 0.15. Success rates remain statistically indistinguishable (99.6% vs. 98.4%, p=0.306).

5.4 Ablation: Guidance Schedule and Scale

We ablate guidance schedules on Lift and Square at s_{base}=0.01, chosen to amplify potential differentiation (at the recommended s=0.005 for Lift, all schedules yield identical $100\%/0.4\%$). All schedules include a 10% warmup (no guidance for the first 10 denoising steps). Table 5.2 reports success and violation rates for all four schedules (constant, linear, cosine, $\sqrt{\cdot}$) on both tasks.

This result reconciles with our motivation (Section 4.3): while the SNR argument predicts lower guidance quality at noisy early steps, the exponential weight $w(p)$ (Eq. 4.1) already suppresses guidance automatically—$\hat{x}_0$ predicted at early steps typically lies far from any obstacle, pushing $w(p) \approx 0$ by the influence-radius mask. Thus temporal modulation via $s(\tau)$ is redundant with the spatial modulation already in $w(p)$. The constant schedule is recommended as the simplest effective

Table 5.2: Guidance schedule ablation (3 seeds $\times$ 50 eps). All four schedules yield statistically indistinguishable results, indicating TACS is robust to schedule choice.

Task	Schedule	Success%	Violation%
Lift (s=0.01)	Constant	92.7 ± 3.8	0.2 ± 0.0
	Linear	92.7 ± 2.5	0.2 ± 0.0
	Cosine	92.7 ± 3.8	0.2 ± 0.0
	$\sqrt{\cdot}$	92.7 ± 3.8	0.2 ± 0.0
Square (s=0.01)	Constant	89.3 ± 2.5	12.0 ± 2.5
	Linear	87.3 ± 2.5	12.6 ± 2.0
	Cosine	87.3 ± 1.9	12.4 ± 2.9
	$\sqrt{\cdot}$	88.7 ± 0.9	12.2 ± 2.9

choice. We next isolate the guidance-scale dimension s itself, sweeping it over more than an order of magnitude for both TACS and PF. Figure 5.4 plots violation rate against s on Lift and Square (log-x): TACS reaches 0% violation at s=0.02 on Lift, while PF requires s=0.1 for comparable safety—a $5\times$ smaller operating scale. On Square, TACS's best operating point strictly Pareto-dominates every PF setting we swept.

5.5 Analysis: In-Loop Compounding Effect

TACS requires an order-of-magnitude smaller guidance scale than PF for comparable violation reduction (e.g., Lift: TACS s=0.005 vs. PF s=0.05). This is because TACS guidance *compounds* across 100 denoising steps: each step nudges $\hat{x}_0$ slightly, and subsequent steps reinforce and adapt the trajectory. PF applies a single correction to the final trajectory without this compounding effect (Figure 1.1). Figure 5.5 provides direct empirical evidence. We record the Tweedie-predicted EE position $\hat{p}_h$ at every denoising step for N=50 action-chunk predictions on Lift. Vanilla's $\hat{p}_h$ hovers just above the obstacle (mean 0.129 m, already near the safety boundary), whereas TACS progressively pushes predictions outward, accumulating a final clearance gain of 2.07 ± 2.63 mm—consistent across calls and highly significant (paired t-test, $p<10^{-5}$, 76% of calls outward). Small per-step nudges thus compound into a reliable reduction in execution-time violations. Figure 5.2 (§5.2) already shows this downstream effect across all three tasks: vanilla DP enters the obstacle region (red $\times$ markers), while TACS and a suitably tuned PF avoid it.

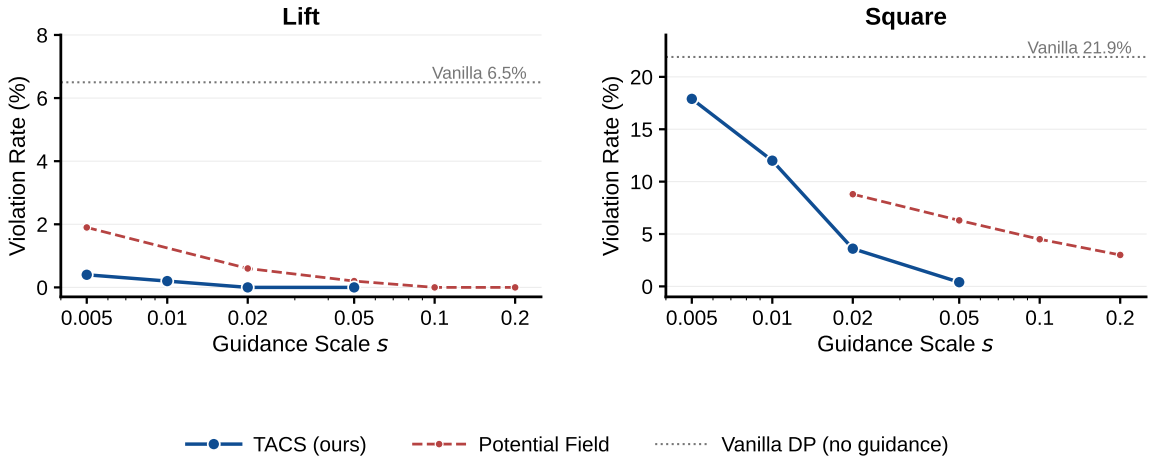


Figure 5.4: **Guidance-scale sensitivity on Lift and Square** (violation rate vs. s, log-x). *Lift:* TACS (blue, solid) reaches 0% violation at s=0.02, whereas PF (red, dashed) needs s=0.1 to match—a $5\times$ smaller guidance scale. *Square:* TACS at s=0.05 achieves 0.4% violation, already below PF's best (3.0% at s=0.2), i.e. PF cannot match TACS at any scale we swept. Vanilla DP (gray, dotted) marks the no-guidance baseline; success rates are in Figure 5.1 and Table 5.1. Mechanism analysis: Section 5.5.

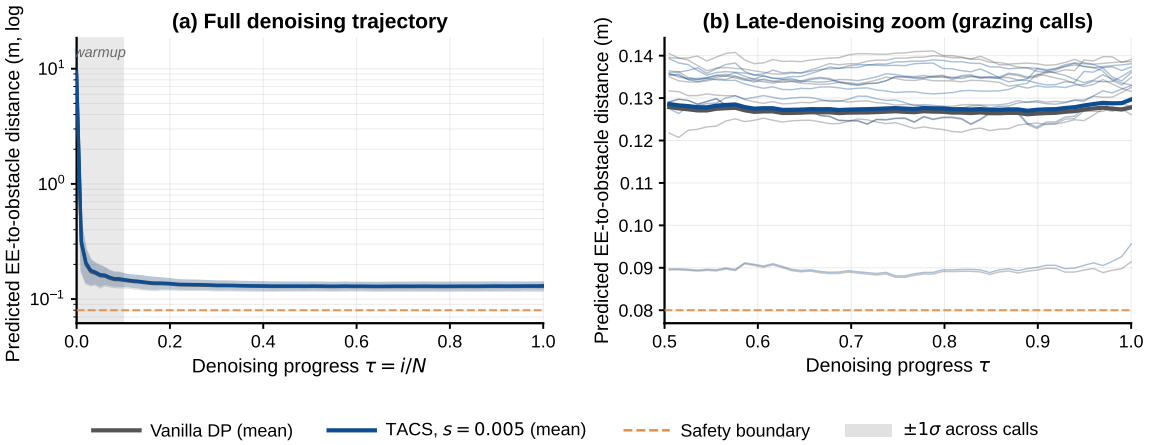


Figure 5.5: **In-loop compounding effect** (N=50 action-chunk predictions on Lift). Distance of the Tweedie-predicted EE $\hat{p}_h$ to the obstacle center versus denoising progress τ. *(a)* Full trajectory (log-y): post-warmup, TACS and Vanilla means are nearly indistinguishable at full scale, motivating the zoom in (b). *(b)* $\tau \in [0.5, 1.0]$ on grazing calls: TACS's per-step nudges accumulate into a $\sim$2.1 mm outward bias over vanilla's $\sim$0.129 m baseline (paired t-test, $p<10^{-5}$). This per-step compounding is the mechanism behind TACS's order-of-magnitude smaller operating s relative to single-shot PF (cf. Figure 1.1, Figure 5.4).

Chapter 6

Conclusion

We presented TACS, a training-free method for enforcing safety constraints in action-space diffusion policies via Tweedie-adaptive barrier guidance within the DDIM denoising loop. Unlike prior inference-time safety methods that target state-space trajectory planners [Xiao et al.(2025), Ma et al.(2025), SafeFlow(2025)], TACS operates directly on the high-dimensional action chunks generated by modern diffusion policies [Chi et al.(2023)], guiding only end-effector position dimensions while allowing the model to coherently adapt rotation and gripper actions. On three robosuite manipulation tasks, TACS achieves consistently better success-violation trade-offs than post-hoc projection, with a statistically significant win-win on Lift (p=0.015) and negligible computational overhead. Ablation studies show that TACS is robust to the guidance schedule choice, with the simple constant schedule performing as well as adaptive alternatives. TACS provides a practical, drop-in safety layer for the increasingly prevalent diffusion policy paradigm.

6.1 Limitations

Experimental scope. We evaluate with single spherical obstacles at fixed positions on three simulation tasks; extending to multiple obstacles, complex geometries, and randomized placements is straightforward but not yet validated. Experiments are simulation-only; real-robot deployment requires perception-based obstacle detection. TACS provides soft guidance without formal safety guarantees; complementing it with a hard safety layer [Dalal et al.(2018)] for deployment-critical scenarios is a natural extension.

Action-space scope. TACS in its current form requires absolute EE-position action parameterization so that the barrier $B_j(\hat{p}_h)$ can be evaluated directly on the Tweedie prediction $\hat{x}_0[h, :3]$. Two extensions are required for broader action spaces: (i) for delta-action policies ($a_h = \Delta p_h$), one must integrate $\hat{p}_h = p_0 + \sum_{k \le h} \Delta \hat{p}_k$ before evaluating B_j, introducing error accumulation across horizon steps; (ii) for joint-space policies ($a_h = q_h$), barrier evaluation requires a differentiable forward-kinematics map $\text{FK}(q_h) \mapsto p_h$, which adds $O(HM \cdot \text{dof})$ Jacobian cost per denoising step. Both extensions are compatible with the TACS framework but are not evaluated in this paper.

Relation to concurrent work. CoDiG [Ma et al.(2025)] applies the same principle of constraint-guided denoising to 2D state-space trajectories for autonomous racing. TACS addresses the distinct challenges of action-space policies (partial-dimension guidance, global observation conditioning), but shares the limitation that guidance strength requires task-specific tuning. Promising future directions include learned safety critics replacing analytic barriers and dynamic obstacle tracking via perception updates between inference calls.

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